Artificial Intelligence: An introduction

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Información del profesor

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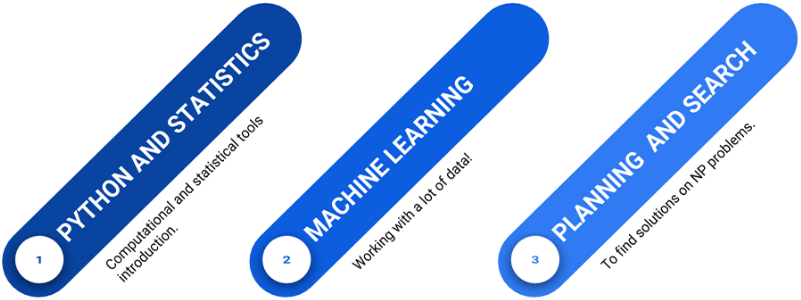
- Contenido del curso: `https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student`

Course outline



- **The course** is mainly based on the **coding of I.A concepts** and the **development of small applications**
- **The course** is focused on the analysis and study of **I.A algorithms**

I.A. Course

Course

It is about learning the fundamental tools to face problems and challenges of I.A. as well as some computational tools and theoretical concepts.

First module: Python and statistics introduction

- Introduction to Python
- Introduction to statistics
- Introduction to Visualization and data analytics

Second Module: Machine Learning

Clasification and regression

- Naive Bayes
- Classification trees
- Support Vector Machines
- Introduction to Deep-learning

Non-supervised learning

- K-means
- DBScan

Third module: Planning and Searching

- Planning and searching
- Genetic algorithms (GA)
- Simulated annealing (SA)

Structure of the Course

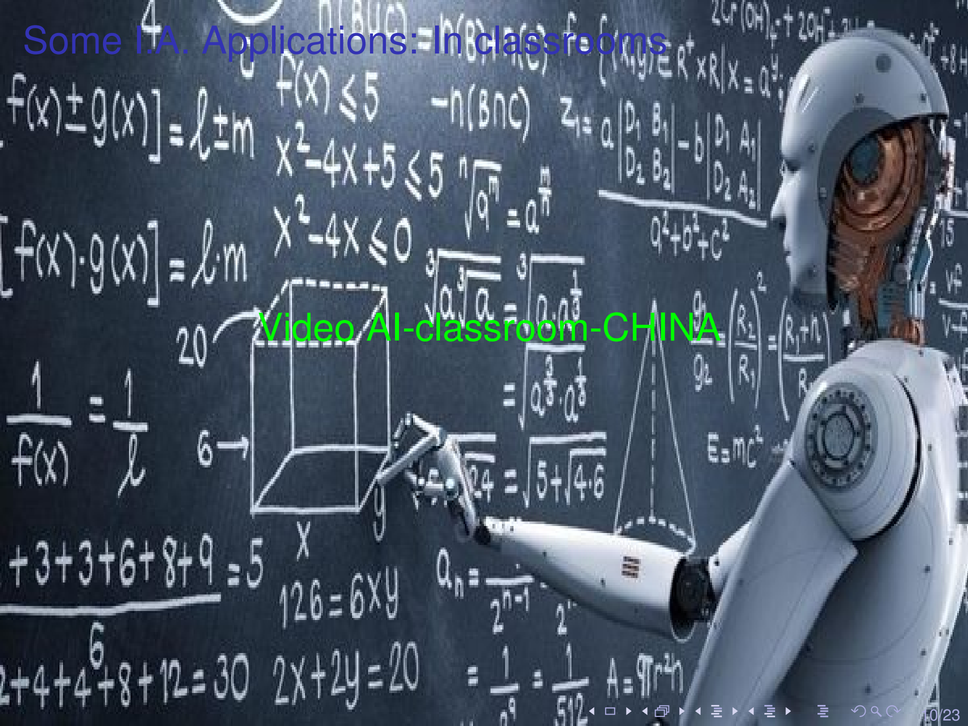
- 30% Talleres (Problemsets)
- 10% Ejercicios clases.
- 30% Parciales (Quizes)
- 30% Proyecto funcional IA.
  - 5 % primera entrega. Exploración de datos
  - 5 % segunda entrega. Validación de modelos clase
  - 20 % Entrega final. Incluye repositorio, presentación y avance del proyecto

Curso en GITLAB:
`https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student`

PREVIOUS PROJECTS

- https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2018-2.md
- https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-1.md
- https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-2.md
  https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2020-1.md

Some I.A. Applications: In classrooms

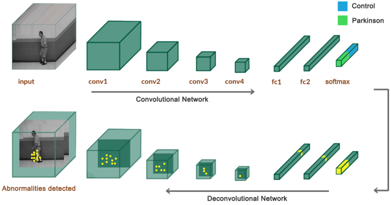


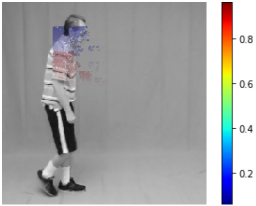
Video AI-classroom-CHINA

Some I.A. Applications: In Agriculture

Video AI-AGRICULTURE

Salient Gait maps

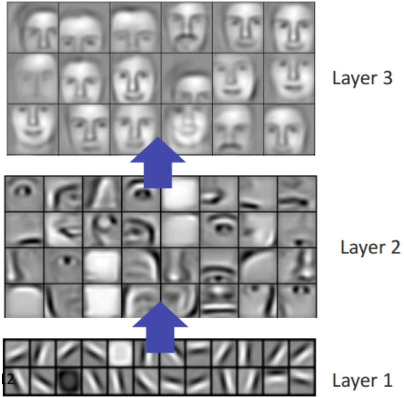


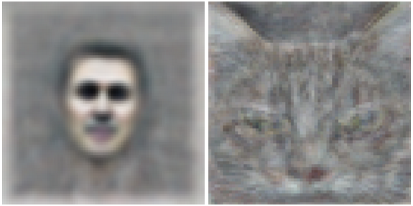


Learning convolutional neuronal patterns

A 3D convolutional gait representation that produces salient spatio-temporal regions can potentially support and complement the diagnosis and the following of Parkinson's disease

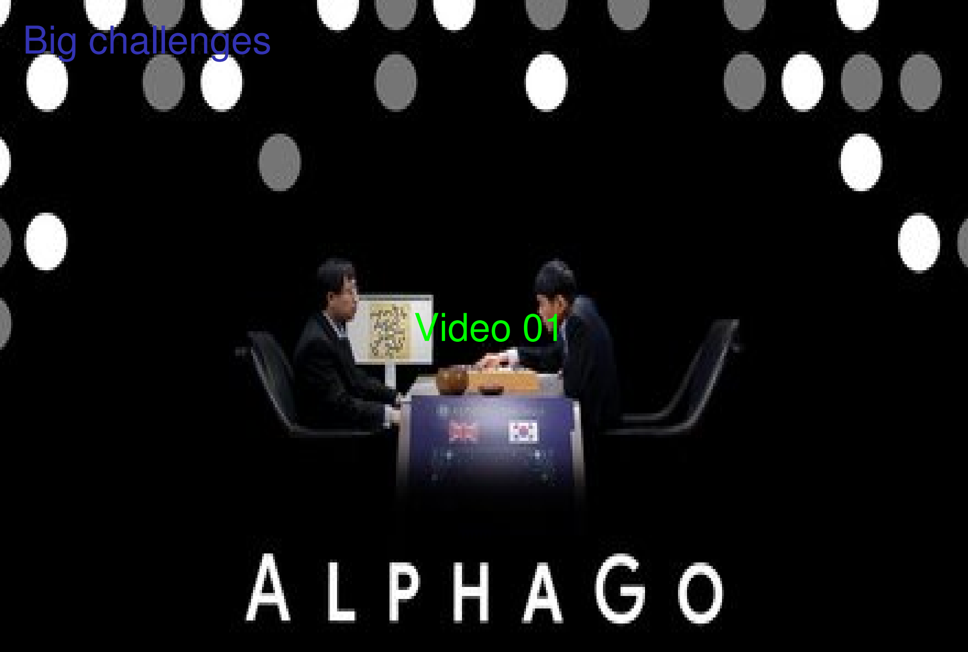
Big challenges: How understand complex concepts?





Now, the machines can dream

Big challenges



Big challenges

ImageNet Challenge

IMAGENET

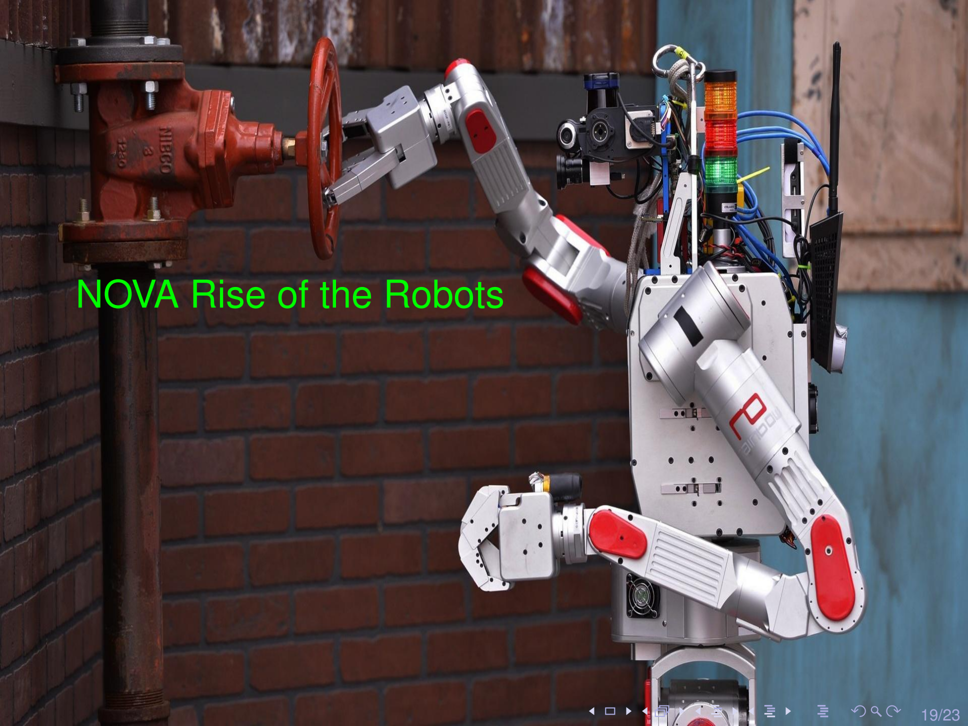
IMAGENET Netflix 22:30. Bill Nye

- 1,000 object classes (categories).
- Images:
  - 1.2 M train
  - 100k test.

As support for decisions making

- Medicine
- Politics
- Laws

NOVA Rise of the Robots



y ahora ...Atlas. Google again!

DARPA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ANDROID
Drivers not required.
SICK

Homework videos

Understanding Artificial Intelligence and Its Future

Thank you for your attention . . .

. . . It’s time to wake up

Universidad Industrial de Santander

MACV
Motion Analysis and Computer Vision

BIVL²ab
Biomedical Imaging, Vision and Learning Laboratory